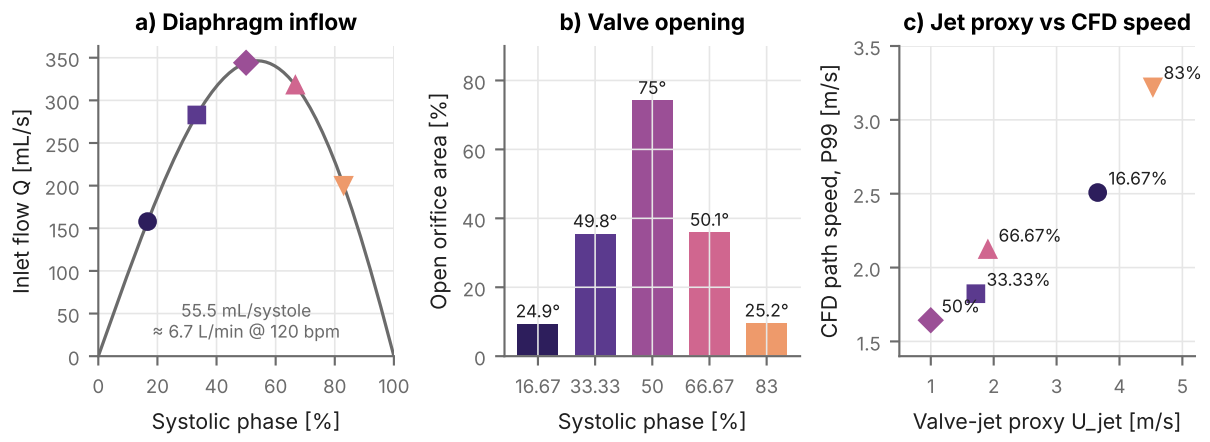
a) Diaphragm inflow
Inlet flow Q [mL/s]
350
300
250
200
150
100
50
0
55.5 mL/systole
≈ 6.7 L/min @ 120 bpm
0
20
40
60
80
100
Systolic phase [%]
b) Valve opening
Open orifice area [%]
80
60
40
20
0
24.9°
49.8°
75°
50.1°
25.2°
16.67
33.33
50
66.67
83
Systolic phase [%]
c) Jet proxy vs CFD speed
CFD path speed, P99 [m/s]
3.5
3.0
2.5
2.0
1.5
83%
16.67%
66.67%
33.33%
50%
1
2
3
4
5
Valve-jet proxy U_jet [m/s]